

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460215.7388 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1603 +/- 0.0041
Transit depth (Rp/Rs)^2	2.57 +/- 0.13 [%]
Orbital Inclination (inc)	86.76 +/- 0.83
Airmass coefficient 1 (a1)	0.982 +/- 0.014
Airmass coefficient 2 (a2)	0.0105 +/- 0.0045
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	#3 - [494, 101]
Optimal Aperture	3.23
Optimal Annulus	14.44
Transit Duration (day)	0.0912 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1603 ± 0.0041 vs 0.1326 (deviation 5.69σ)
Duration fit vs published	0.0912 d vs 0.0737 d (deviation 3.13σ)
Mid-time O–C	-3.6 min
Depth SNR	32.9
Residual scatter	1.41 %

Light curve

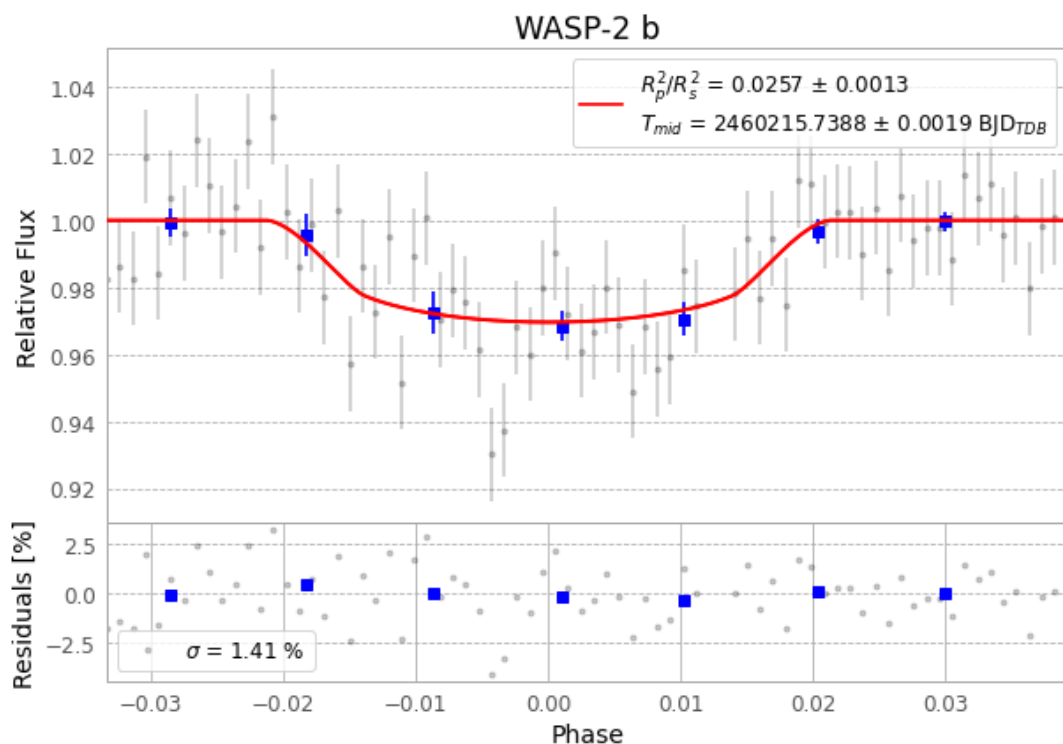


Figure 1 — FinalLightCurve_WASP-2 b_2023-09-27.png (SHA-256 a160e735a7837f54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a033777e9b7e34c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1cb2214

Data provenance

76 FITS frames (50 MB), WASP-2230928035415.FITS → WASP-2230928073915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230928.log (26146e5c765b...), FinalLightCurve_WASP-2 b_2023-09-27.png (a160e735a783...), FinalLightCurve_WASP-2 b_2023-09-27.csv (7a8ff50f44f8...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 12:55Z.